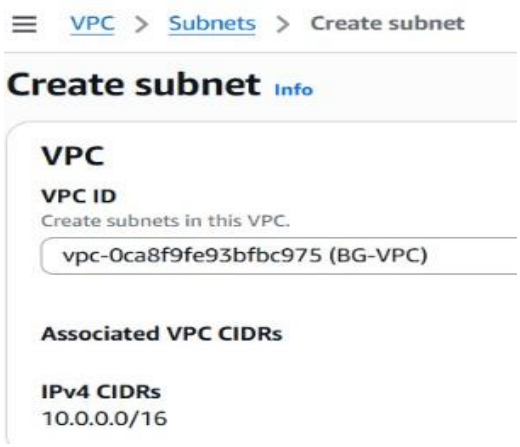
Blue/Green Deployment in AWS

Step 1: Create a VPC

1. Open AWS Console.
2. Search for VPC.
3. Click Create VPC.
4. Name: BG-VPC
5. IPv4 CIDR: `10.0.0.0/16`
6. Click Create VPC.

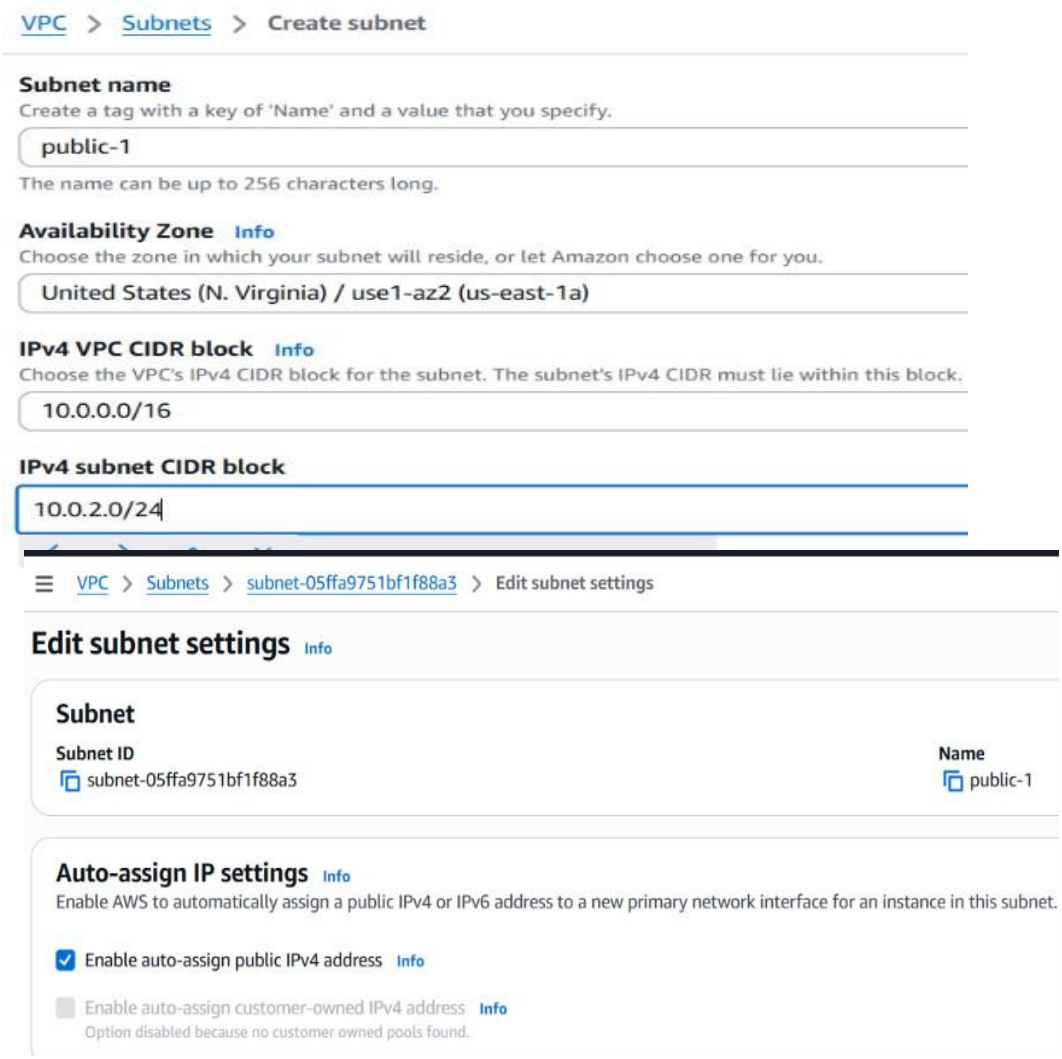
Screenshot 1: BG-VPC created successfully



Step 2: Create Public Subnet 1

1. Open VPC → Subnets → Create subnet.
2. Select VPC: `BG-VPC`
3. Subnet name: `Public-1`
4. Availability Zone: `us-east-1a`
5. IPv4 subnet CIDR block: `10.0.2.0/24`
6. Click Create subnet.
7. Select the subnet and click Actions → Edit subnet settings.
8. Enable Auto-assign public IPv4 address.
9. Save the settings.

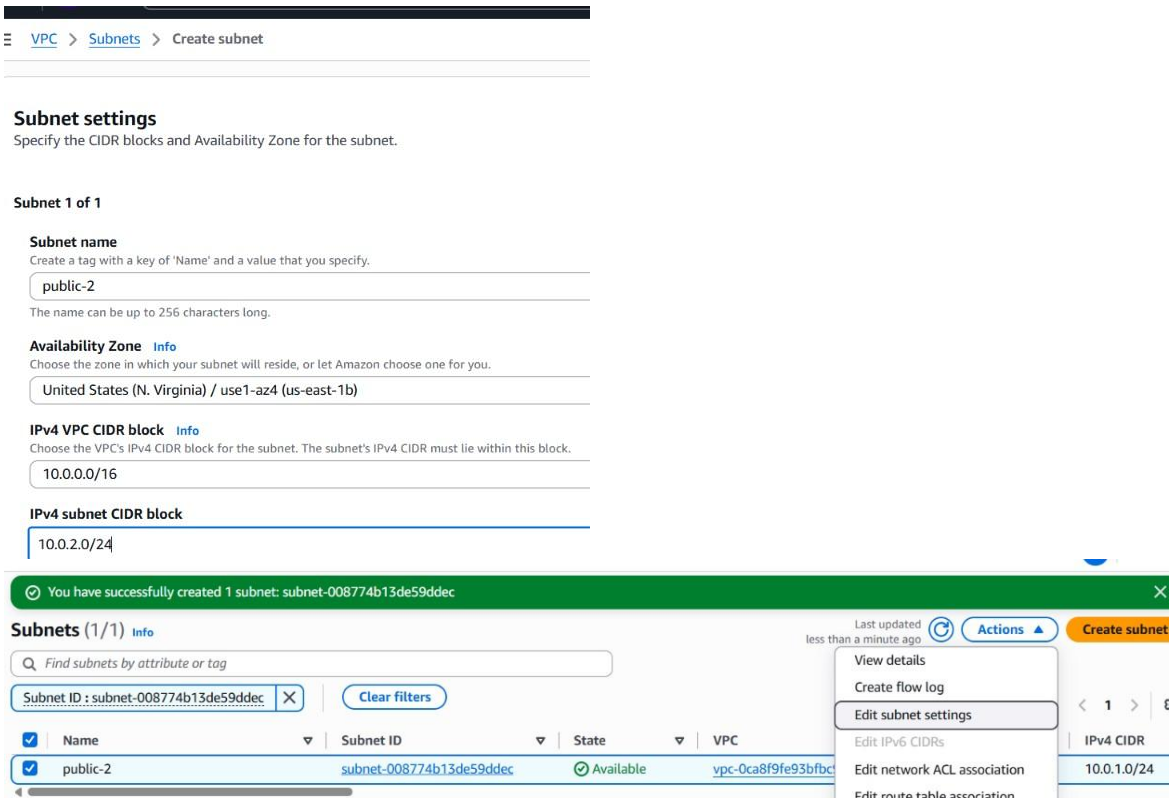
Screenshot 2: Public-1 with its CIDR and Auto-assign public IPv4 enabled.



Step 3: Create Public Subnet 2

1. Click Create subnet.
2. Select VPC: `BG-VPC`.
3. Subnet name: `Public-2`.
4. Availability Zone: `us-east-1b`.
5. IPv4 subnet CIDR block: `10.0.3.0/24`.
6. Create the subnet.
7. Open Edit subnet settings.
8. Enable Auto-assign public IPv4 address.
9. Save the settings.

Screenshot 3: Public-1 and Public-2 in different Availability Zones.



Step 4: Create an Internet Gateway

1. Open VPC → Internet gateways.
2. Click Create internet gateway.
3. Name: `BG-IGW`.
4. Click Create internet gateway.
5. Select `BG-IGW`.
6. Click Actions → Attach to a VPC.
7. Select `BG-VPC`.
8. Click Attach internet gateway.

Screenshot 4: Internet Gateway attached to BG-VPC.



Step 5: Create a Route Table

1. Open VPC → Route tables.
2. Click Create route table.

3. Name: `BG-Rt`.
4. Select VPC: `BG-VPC`.
5. Click Create route table.

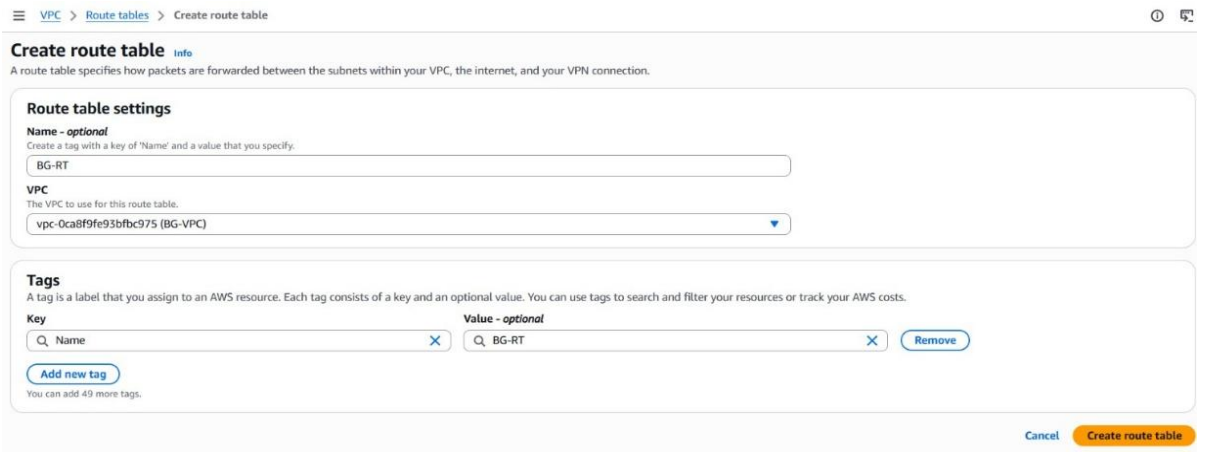
Add an Internet route

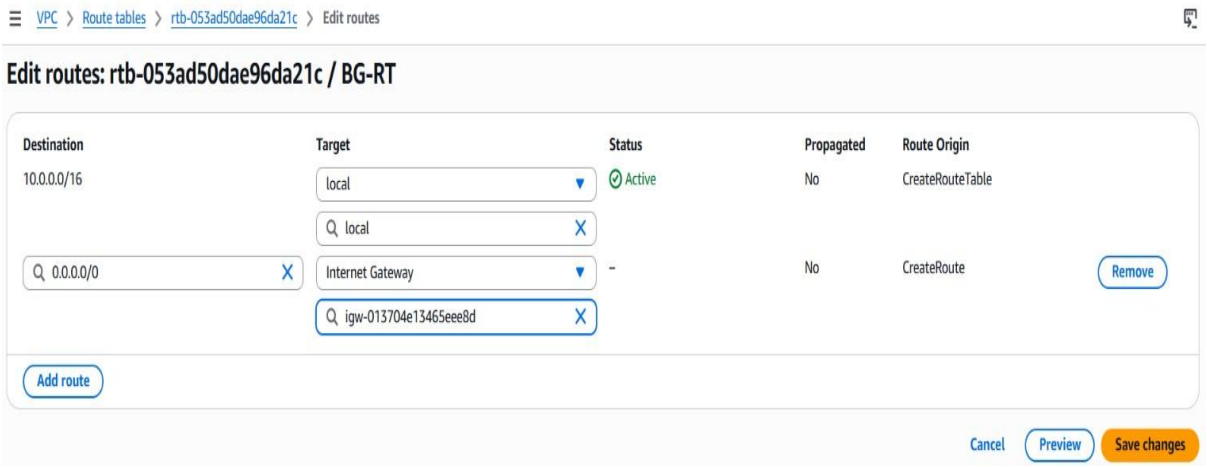
1. Select `BG-Rt`.
2. Open the Routes tab.
3. Click Edit routes → Add route.
4. Destination: `0.0.0.0/0`
5. Target: Select Internet Gateway → BG-IGW.
6. Click Save changes.

Associate both public subnets

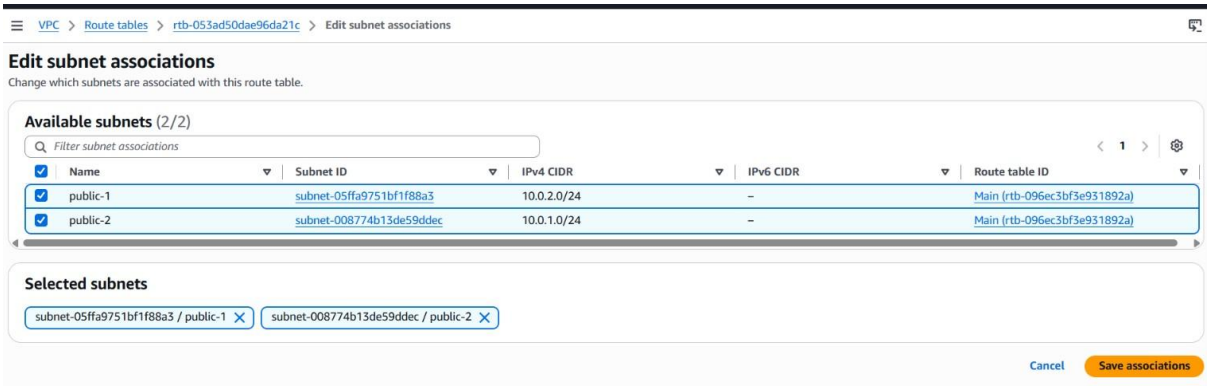
1. Open the Subnet associations tab.
2. Click Edit subnet associations.
3. Select `Public-1` and `Public-2`.
4. Click Save associations.

Screenshot 5: Route table showing `0.0.0.0/0` pointing to BG-IGW.





Screenshot 6: Both public subnets associated with BG-Rt.



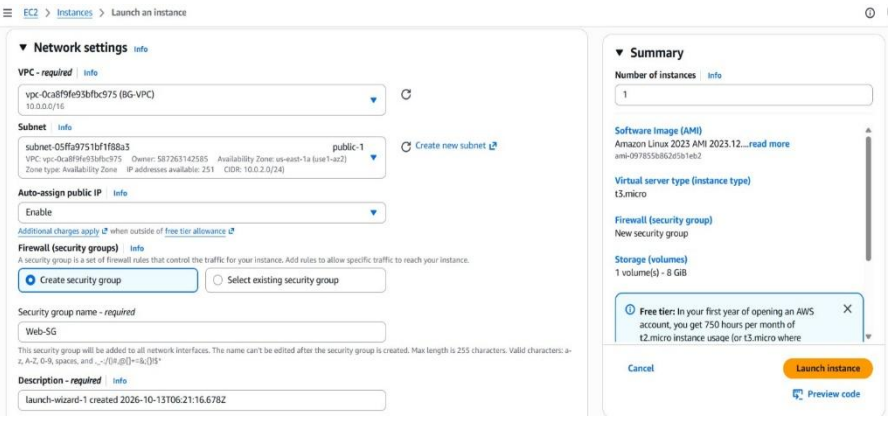
Step 6: Create a Security Group

1. Open EC2 → Security Groups.
2. Click Create security group.
3. Security group name: `web-sg`
4. Description: `Allow web traffic`
5. VPC: `BG-VPC`
6. Add inbound rules:

Type	Port	Source
HTTP	80	`0.0.0.0/0`
SSH	22	Your IP only, if required

7. Click Create security group.

Screenshot 7: Security group `web-sg` with the HTTP rule.



Step 7: Launch the Blue EC2 Instance

1. Open EC2 → Instances → Launch instances.
2. Name: `Blue-Server`
3. Select the Amazon Linux AMI specified by the lab.
4. Select the instance type specified by the lab.

5. Select or create a key pair if required.
6. Under Network settings, click Edit:
    - VPC: `BG-VPC`
    - Subnet: `Public-1`
    - Auto-assign public IP: Enable
    - Security group: `web-sg`

8. Expand **Advanced details → User data**.
9. Paste the following script into the User data field.
10. Click **Launch instance**.

```
#!/bin/bash

yum update -y

yum install -y httpd

systemctl start httpd

systemctl enable httpd



cat > /var/www/html/index.html <<'EOF'

<html>

<body style="background-color:blue;color:white;">

<h1>Version 1 - Blue</h1>

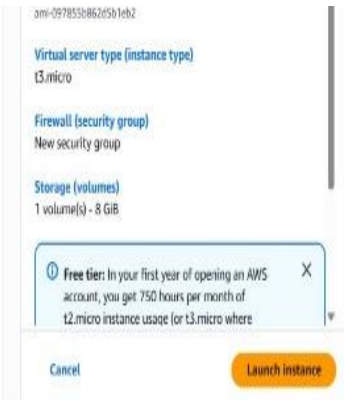
</body>

</html>

EOF
```

Screenshot 8: Blue-Server running coding





Step 8: Launch the Green EC2 Instance

1. Click Launch instances.
2. Name: `Green-Server`.
3. Select the same AMI and instance type as Blue.
4. Select VPC: `BG-VPC`.
5. Select Subnet: `Public-2`.
6. Enable Auto-assign public IP.
7. Select security group: `web-sg`.
8. Expand Advanced details → User data.
9. Paste the following script:

```
#!/bin/bash

yum update -y

yum install -y httpd

systemctl start httpd

systemctl enable httpd

cat > /var/www/html/index.html <<'EOF'

<html>

<body style="background-color:green;color:white;">

<h1>Version 2 - Green</h1>

</body>

</html>

EOF
```

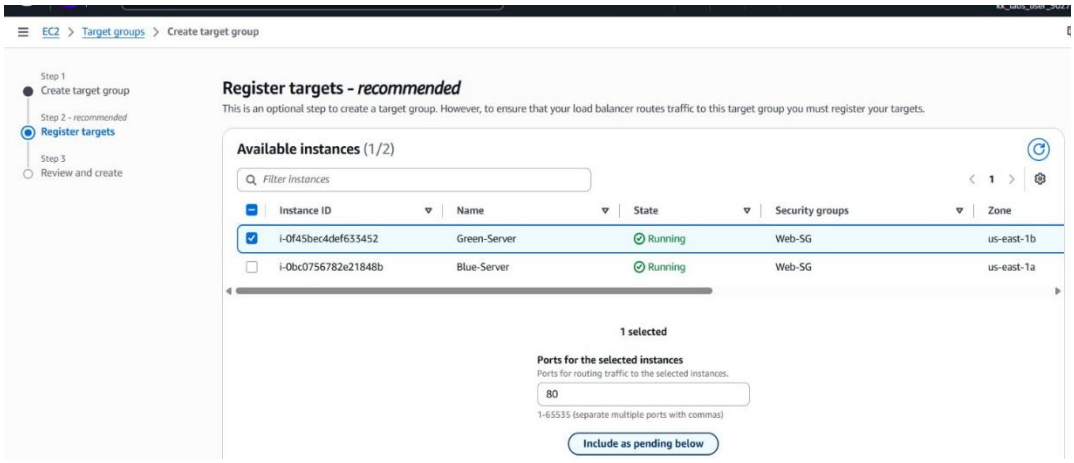
10. Click Launch instance.

Step 9: Create the Blue Target Group

1. Open EC2 → Target Groups.
2. Click Create target group.
3. Target type: Instances.
4. Target group name: `BG-Blue-TG`.
5. Protocol: `HTTP`.
6. Port: `80`.
7. VPC: `BG-VPC`.
8. Health check protocol: `HTTP`.
9. Health check path: `/`.

10. Click Next.
11. Select `Blue-Server`.
12. Click Include as pending below.
13. Click Create target group.

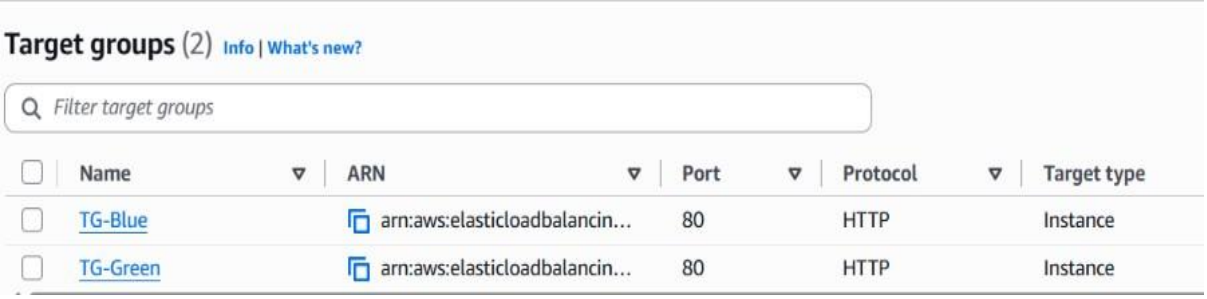
Screenshot 10: BG-Blue-TG with Blue-Server registered.



Step 10: Create the Green Target Group

1. Open EC2 → Target Groups → Create target group.
2. Target type: Instances.
3. Target group name: `BG-Green-TG`.
4. Protocol: `HTTP`.
5. Port: `80`.
6. VPC: `BG-VPC`.
7. Health check path: `/`.
8. Click Next.
9. Select `Green-Server`.
10. Click Include as pending below.
11. Click Create target group.

Screenshot 11: BG-Green-TG with Green-Server registered.

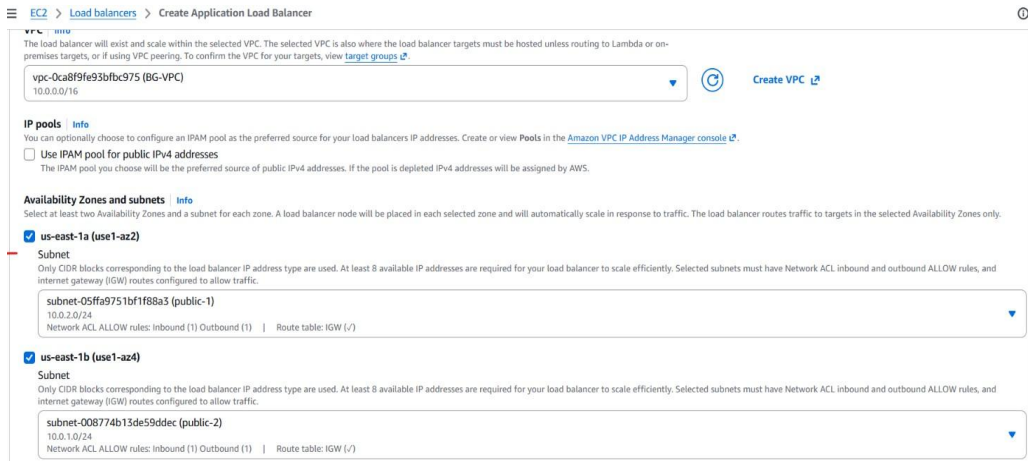


Step 11: Create an Application Load Balancer

1. Open EC2 → Load Balancers.
2. Click Create load balancer.

3. Select Application Load Balancer.
4. Name: `BG-ALB`.
5. Scheme: Internet-facing.
6. IP address type: IPv4.
7. Select VPC: `BG-VPC`.
8. Under Network mapping, select:
    - Availability Zone `us-east-1a` → `Public-1`
    - Availability Zone `us-east-1b` → `Public-2`
9. Select security group: `web-sg`.
10. Under Listeners and routing, choose HTTP port `80`.
11. Select the default target group `BG-Blue-TG`.
12. Click Create load balancer.

Screenshot 12: ALB configuration with both public subnets selected.



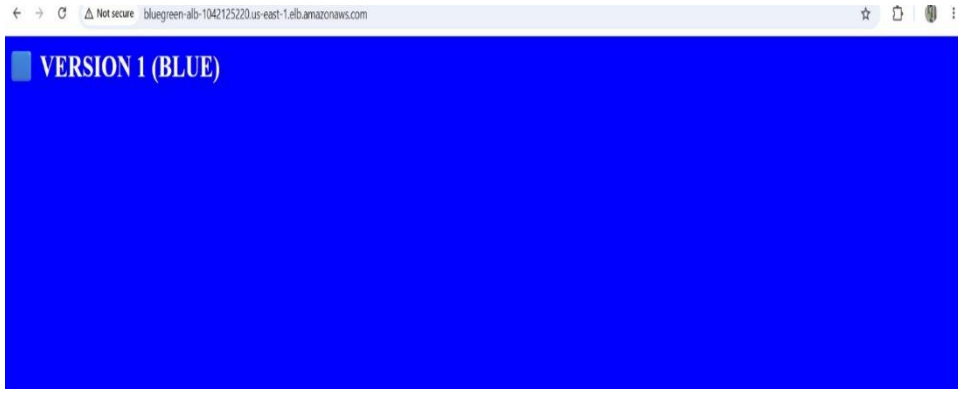
1. Open EC2 → Load Balancers.
2. Select `BG-ALB`.
3. Copy its DNS name.
4. Open the DNS name in a browser using HTTP.

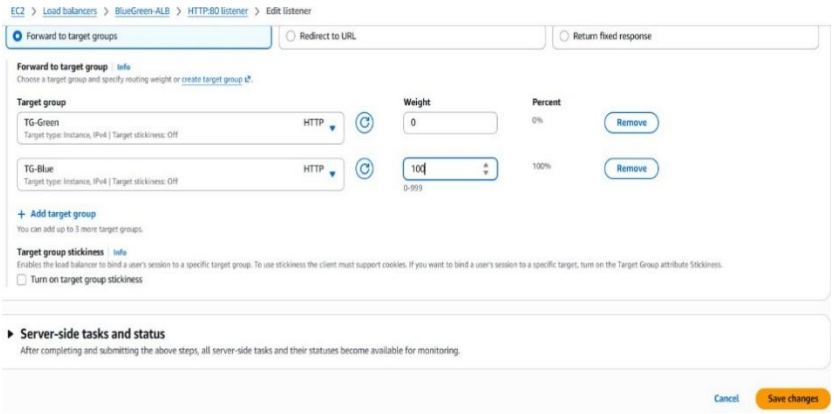
Example URL format:

```
http://your-load-balancer-dns-name
```

You should see Version 1 - Blue if the Blue target is healthy and the listener forwards traffic to it.

Screenshot 13: Blue website opened through the ALB DNS name.





Step 13: Switch Traffic from Blue to Green

1. Open EC2 → Load Balancers.
2. Select `BG-ALB`.
3. Open the Listeners and rules tab.
4. Select the HTTP port `80` listener.
5. Click Manage rules or Edit listener, depending on the console screen.
6. Edit the default action.
7. Change the forwarding target from `BG-Blue-TG` to `BG-Green-TG`.
8. Save the changes.

Step 14: Verify the Green Version

1. Open the same ALB DNS name again.
2. Refresh the browser.
3. Check the application page.
4. You should see Version 2 - Green.

Screenshot 14: Green website opened through the same ALB DNS name.



Final project summary

Project: Blue-Green Deployment on AWS

Blue Environment: Version 1 — Existing application

Green Environment: Version 2 — New application

Infrastructure: Amazon VPC, Public Subnets, Internet Gateway, Route Table, EC2, Security Groups, Target Groups, and Application Load Balancer.

Conclusion: Blue-Green Deployment enables safer application releases by maintaining separate Blue and Green environments. Traffic can be switched through the Application Load Balancer after validating the new version. If required, traffic can be switched back to the previous environment, reducing deployment risk and downtime.

Cost Reminder: Delete unused EC2 instances, load balancers, and other billable resources after completing the lab.